



AI reinvention made real

Accenture Innovation Challenge 2026

Team Details

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Enterprise GenAI Reality

THE SILENT DEGRADATION TRIAD



1. PERFORMANCE RISK

Invalid answers delivered with false confidence



Confident Hallucinations
Models state plausible but false information with high certainty, masking errors until it's too late.



Context Drift & Inversion
Physical or logical details are altered—changing meaning and outcomes.



Entity & Numeric Flips
Swap names, numbers or entities (e.g., \$1.5% vs \$15%) in contracts, reports or legal summaries.



Confidence Masking
Errors appear with flat, confident tone—users act on them without realizing.



IMPACT: Bad decisions, compliance breaches, customer harm, brand erosion.



2. COST / COMPUTE RISK

Silent token hemorrhage at scale



Redundant Query Waste
Thousands ask the same question daily. Without semantic caching, every query hits expensive models—millions wasted.



Model Oversizing
Up to 70% of tasks are simple—yet routed to \$100B+ models when a small model would suffice.



Runaway Agentic Loops
Agents get stuck in retry loops, burning tens of thousands of tokens without useful output.



Escalating Cloud Spend
Verbose prompts, long outputs and inefficient retrieval inflate tokens, latency and cost.



IMPACT: Millions in wasted spend, poor ROI and budget overruns.



3. RESPONSIBILITY RISK

Sensitive data exposure & adversarial ingress



In-Flight PII/PHI Leaks
SSNs, MRNs, cards, API keys and other sensitive data leak across raw streaming outputs and logs.



Indirect Prompt Injections
Poisoned docs or inputs override instructions and extract system prompts or data.



Harmful / Unsafe Outputs
Biased, toxic or illegal content can be generated without guardrails.



Regulatory & Legal Exposure
EU AI Act, HIPAA, GDPR violations—fines up to €35M or 7% of global turnover.



IMPACT: Regulatory fines, lawsuits, data breaches and loss of trust.



THE CORE INDUSTRY NEED

A technology layer that provides in-flight, real-time token-level observation and intervention with sub-millisecond wire-speed performance—catching and resolving risks **before** tokens are displayed.

THE STAKES OF SILENCE



\$MILLIONS

Wasted on redundant tokens & model misallocation



INCORRECT ACTIONS

Drive bad business decisions, compliance breaches, & customer harm



LOSS OF TRUST

Erodes user confidence and brand credibility at scale



REGULATORY FINES

Up to €35M or 7% of global turnover under EU AI Act / GDPR / HIPAA

THE PRODUCTION GOVERNANCE VACUUM

APPROACH A: POST-HOC BATCH EVALS

Mechanism: Asynchronous log parsing & evals run hours or days after request completion.

Fatal Flaw: Catching an error after 24 hours means the user has already received and acted on wrong or dangerous information.

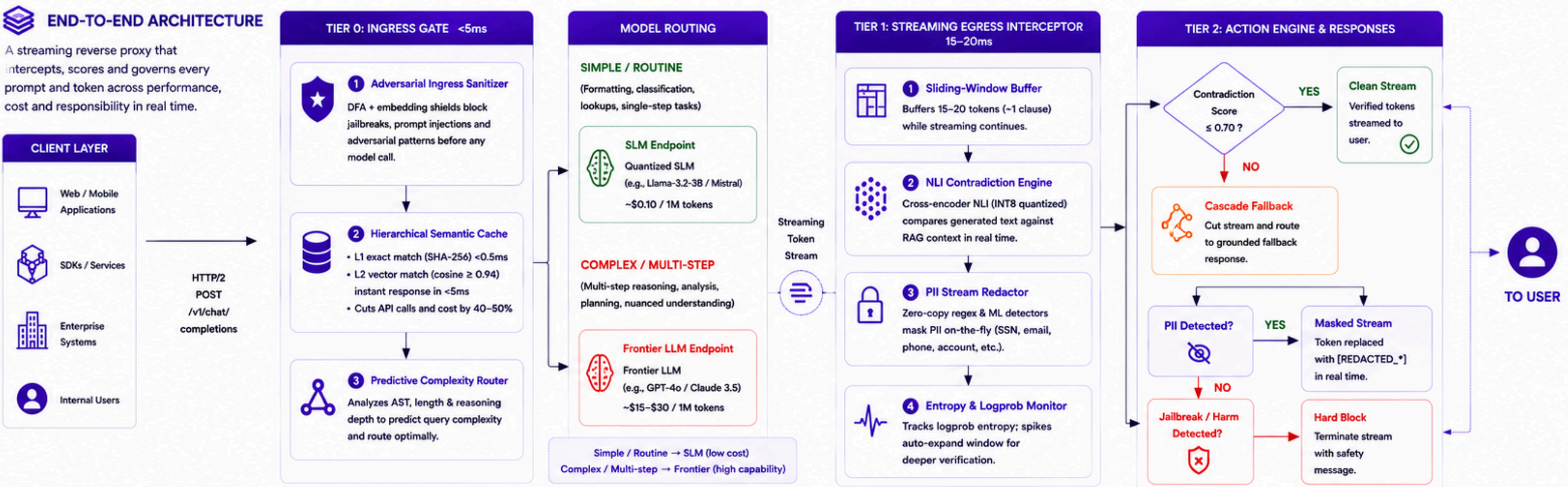
APPROACH B: INLINE LLM-AS-A-JUDGE

Mechanism: Synchronously call a secondary LLM on every output to verify safety & accuracy.

Fatal Flaw: Adds 1,200ms—2,500ms of latency; doubles API costs. Destroys real-time streaming conversational UX.

BOTH APPROACHES LEAVE A CRITICAL GAP:
No in-flight, real-time token-level intervention.

Control Plane AI : Real time governance for every prompt and token



CONTROLPLANE.AI COMPLETE TECH STACK					
LAYER		PRODUCTION TECHNOLOGY		EXACT TECHNICAL ROLE & PERFORMANCE SLA	KEY METRICS
1	 Frontend Interface	 Streamlit (Python)		Interactive Enterprise Command Center: Live Prompt Playground, Streaming Token Inspector, Policy Configurator, 44 Judge Scenarios.	 TTFT Overhead < 20 ms
2	 Streaming Gateway	 Flask / FastAPI (Async)		Model-agnostic reverse proxy intercepting OpenAI/Bedrock Server-Sent Events (SSE) deltas over HTTP/2 with zero client code changes.	 Supports 10K+ Concurrent Streams
3	 Inference Runtime	 ONNX Runtime (INT8) / TensorRT-LLM		Hardware-accelerated local execution of distilled evaluators yielding sub-15ms NLI cross-encoder checks without cloud calls.	 NLI Inference < 15 ms
4	 Grounding & NLI Model	DeBERTa-v3-small-NLI (INT8) / ModernBERT Cross-Encoder		Evaluates premise-hypothesis factual entailment against retrieved RAG chunks in real time (88M parameters, <90MB memory footprint).	 Accuracy (NLI) > 94% F1
5	 Semantic Vector Cache	Redis Vector Search / Qdrant (all-MiniLM-L6-v2)	 	In-memory HNSW index on 384-dim embeddings (Cosine Sim ≥ 0.94); returns cached answers in <5ms, bypassing upstream LLM calls.	 Cache Hit Latency < 5 ms
6	 Observability & Audit	 OpenTelemetry + ClickHouse + Apache Kafka	 	Asynchronous, non-blocking telemetry emitter storing immutable per-token audit trails for EU AI Act (High-Risk Systems) & HIPAA audits.	 Durable Audit Logs 100% Immutable

WHY IT MATTERS: FOUR ENTERPRISE IMPACT PILLARS

1 Factual Integrity & Hallucination Elimination



A single hallucinated figure can cause millions in financial loss or severe liability.

- Continuous NLI checks on streaming clauses detect ungrounded assertions and truncate the stream before falsehoods reach users.

2 Unit Economics & Compute Cost Optimization



Enterprises overspend due to repeated queries and oversized models.

- In-memory vector caching cuts LLM calls by 40–50%.
- Predictive routing sends simple queries to SLMs (e.g., Llama-3.2-3B at \$0.10/1M tokens).

3 Human-Grade Streaming Speed (Zero Latency Penalty)



Traditional guardrails add 1,200–2,500ms latency—killing conversational UX.

- Speculative sliding-window buffer evaluates 15-token clauses in parallel on local runtimes.
- Safe tokens stream at wire speed with <20ms TTFT overhead.

Goal: <20ms TTFT Overhead

4 Turnkey Regulatory Readiness & Legal Protection



EU AI Act fines up to €35M. HIPAA & GDPR penalize PII/PHI leaks.

- In-stream DFA masks redact PII (SSN, CC, Medical IDs) before output.
- Immutable audit lake logs every token risk score, NLI probability & model version for compliance.

Goal: 100% Auditability



Presentation Video : [Link](#)

THE PARADIGM SHIFT

“ Traditional AI observability is reactive—catching catastrophic hallucinations and data breaches during weekly log reviews, long after users have already acted on them. Inline LLM judges are too slow and double the cost. **ControlPlane.ai transforms AI governance from an after-the-fact post-mortem into real-time, in-flight runtime protection.** ”

OUR COMPETITIVE MOATS



Drop-In Compatibility

Zero application refactoring—change a single base_url endpoint.



Edge-to-Cloud Flexibility

Runs as a sidecar container on Kubernetes, edge proxy, or enterprise gateway.



Model & Cloud Agnostic

Protects OpenAI, Anthropic, Bedrock, Azure, or self-hosted (vLLM/Triton) models.

Thank you

